MSG No. 142.562 $I4_1/acd1'$ [Type II, tetragonal]

Table 1: Wyckoff site: `8a`, site symmetry: `-4..1'`

No.	position	mapping
1	$\left[0, \frac{1}{4}, \frac{3}{8}\right]$	`[1,6,10,11,33,38,42,43]`
2	$\left[0, \frac{3}{4}, \frac{5}{8}\right]$	`[2,3,9,14,34,35,41,46]`
3	$\left[\frac{1}{2}, \frac{1}{4}, \frac{5}{8}\right]$	`[4,5,15,16,36,37,47,48]`
4	$\left[0, \frac{1}{4}, \frac{7}{8}\right]$	`[7,8,12,13,39,40,44,45]`
5	$\left[\frac{1}{2}, \frac{3}{4}, \frac{7}{8}\right]$	`[17,22,26,27,49,54,58,59]`
6	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{8}\right]$	`[18,19,25,30,50,51,57,62]`
7	$\left[0, \frac{3}{4}, \frac{1}{8}\right]$	`[20,21,31,32,52,53,63,64]`
8	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{8}\right]$	`[23,24,28,29,55,56,60,61]`

Table 2: Wyckoff site: `8b`, site symmetry: `2.221'`

No.	position	mapping
1	$\left[0, \frac{1}{4}, \frac{1}{8}\right]$	`[1,6,7,8,33,38,39,40]`
2	$\left[0, \frac{3}{4}, \frac{3}{8}\right]$	`[2,3,20,21,34,35,52,53]`
3	$\left[\frac{1}{2}, \frac{1}{4}, \frac{7}{8}\right]$	`[4,5,18,19,36,37,50,51]`
4	$\left[0, \frac{3}{4}, \frac{7}{8}\right]$	`[9,14,31,32,41,46,63,64]`
5	$\left[0, \frac{1}{4}, \frac{5}{8}\right]$	`[10,11,12,13,42,43,44,45]`
6	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{8}\right]$	`[15,16,25,30,47,48,57,62]`
7	$\left[\frac{1}{2}, \frac{3}{4}, \frac{5}{8}\right]$	`[17,22,23,24,49,54,55,56]`
8	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{8}\right]$	`[26,27,28,29,58,59,60,61]`

Table 3: Wyckoff site: `16c`, site symmetry: `-11'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,9,33,41]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[2,26,34,58]`
3	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[3,27,35,59]`
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[4,28,36,60]`
5	$\left[\frac{1}{2}, 0, 0\right]$	`[5,29,37,61]`
6	$\left[0, \frac{1}{2}, 0\right]$	`[6,14,38,46]`
7	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[7,15,39,47]`
8	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[8,16,40,48]`
9	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[10,18,42,50]`
10	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[11,19,43,51]`
11	$\left[0, 0, \frac{1}{2}\right]$	`[12,20,44,52]`
12	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[13,21,45,53]`
13	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[17,25,49,57]`
14	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[22,30,54,62]`
15	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[23,31,55,63]`
16	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[24,32,56,64]`

Table 4: Wyckoff site: `16d`, site symmetry: `2..1'`

No.	position	mapping
1	$\left[0,\ \frac{1}{4},\ z\right]$	`[1,6,33,38]`
2	$\left[0,\ \frac{3}{4},\ z+\frac{1}{4}\right]$	`[2,3,34,35]`
3	$\left[\frac{1}{2},\ \frac{1}{4},\ -z\right]$	`[4,5,36,37]`
4	$\left[0,\ \frac{1}{4},\ \frac{1}{4}-z\right]$	`[7,8,39,40]`
5	$\left[0,\ \frac{3}{4},\ -z\right]$	`[9,14,41,46]`
6	$\left[0,\ \frac{1}{4},\ \frac{3}{4}-z\right]$	`[10,11,42,43]`
7	$\left[0,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[12,13,44,45]`
8	$\left[\frac{1}{2},\ \frac{1}{4},\ z+\frac{1}{4}\right]$	`[15,16,47,48]`
9	$\left[\frac{1}{2},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[17,22,49,54]`
10	$\left[\frac{1}{2},\ \frac{1}{4},\ z+\frac{3}{4}\right]$	`[18,19,50,51]`
11	$\left[0,\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[20,21,52,53]`
12	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{3}{4}-z\right]$	`[23,24,55,56]`
13	$\left[\frac{1}{2},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	`[25,30,57,62]`
14	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{1}{4}-z\right]$	`[26,27,58,59]`
15	$\left[\frac{1}{2},\ \frac{3}{4},\ z\right]$	`[28,29,60,61]`
16	$\left[0,\ \frac{3}{4},\ z+\frac{3}{4}\right]$	`[31,32,63,64]`

Table 5: Wyckoff site: `16e`, site symmetry: `.2.1'`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,20,33,52]`
2	$\left[\frac{1}{4},\ x+\frac{3}{4},\ \frac{1}{2}\right]$	`[2,23,34,55]`
3	$\left[\frac{3}{4},\ \frac{3}{4}-x,\ \frac{1}{2}\right]$	`[3,24,35,56]`
4	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[4,17,36,49]`
5	$\left[\frac{1}{2}-x,\ 0,\ \frac{3}{4}\right]$	`[5,22,37,54]`
6	$\left[-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[6,21,38,53]`
7	$\left[\frac{3}{4},\ x+\frac{1}{4},\ 0\right]$	`[7,18,39,50]`
8	$\left[\frac{1}{4},\ \frac{1}{4}-x,\ 0\right]$	`[8,19,40,51]`
9	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[9,12,41,44]`
10	$\left[\frac{3}{4},\ \frac{1}{4}-x,\ \frac{1}{2}\right]$	`[10,15,42,47]`
11	$\left[\frac{1}{4},\ x+\frac{1}{4},\ \frac{1}{2}\right]$	`[11,16,43,48]`
12	$\left[x,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[13,14,45,46]`
13	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[25,28,57,60]`
14	$\left[\frac{1}{4},\ \frac{3}{4}-x,\ 0\right]$	`[26,31,58,63]`
15	$\left[\frac{3}{4},\ x+\frac{3}{4},\ 0\right]$	`[27,32,59,64]`
16	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[29,30,61,62]`

Table 6: Wyckoff site: `16f`, site symmetry: `..21'`

No.	position	mapping
1	$\left[x,\ x+\frac{1}{4},\ \frac{1}{8}\right]$	`[1,7,33,39]`
2	$\left[-x,\ x+\frac{3}{4},\ \frac{3}{8}\right]$	`[2,21,34,53]`
3	$\left[x,\ \frac{3}{4}-x,\ \frac{3}{8}\right]$	`[3,20,35,52]`
4	$\left[x+\frac{1}{2},\ \frac{1}{4}-x,\ \frac{7}{8}\right]$	`[4,19,36,51]`

continued ...

Table 6

No.	position	mapping
5	$\left[\frac{1}{2}-x,\, x+\frac{1}{4},\, \frac{7}{8}\right]$	[5,18,37,50]
6	$\left[-x,\, \frac{1}{4}-x,\, \frac{1}{8}\right]$	[6,8,38,40]
7	$\left[-x,\, \frac{3}{4}-x,\, \frac{7}{8}\right]$	[9,31,41,63]
8	$\left[x,\, \frac{1}{4}-x,\, \frac{5}{8}\right]$	[10,13,42,45]
9	$\left[-x,\, x+\frac{1}{4},\, \frac{5}{8}\right]$	[11,12,43,44]
10	$\left[x,\, x+\frac{3}{4},\, \frac{7}{8}\right]$	[14,32,46,64]
11	$\left[\frac{1}{2}-x,\, \frac{1}{4}-x,\, \frac{3}{8}\right]$	[15,25,47,57]
12	$\left[x+\frac{1}{2},\, x+\frac{1}{4},\, \frac{3}{8}\right]$	[16,30,48,62]
13	$\left[x+\frac{1}{2},\, x+\frac{3}{4},\, \frac{5}{8}\right]$	[17,23,49,55]
14	$\left[\frac{1}{2}-x,\, \frac{3}{4}-x,\, \frac{5}{8}\right]$	[22,24,54,56]
15	$\left[x+\frac{1}{2},\, \frac{3}{4}-x,\, \frac{1}{8}\right]$	[26,29,58,61]
16	$\left[\frac{1}{2}-x,\, x+\frac{3}{4},\, \frac{1}{8}\right]$	[27,28,59,60]

Table 7: Wyckoff site: 32g, site symmetry: 11'

No.	position	mapping
1	$[x,\, y,\, z]$	[1,33]
2	$\left[\frac{1}{4}-y,\, x+\frac{3}{4},\, z+\frac{1}{4}\right]$	[2,34]
3	$\left[y+\frac{3}{4},\, \frac{3}{4}-x,\, z+\frac{1}{4}\right]$	[3,35]
4	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	[4,36]
5	$\left[\frac{1}{2}-x,\, y,\, -z\right]$	[5,37]
6	$\left[-x,\, \frac{1}{2}-y,\, z\right]$	[6,38]
7	$\left[y+\frac{3}{4},\, x+\frac{1}{4},\, \frac{1}{4}-z\right]$	[7,39]
8	$\left[\frac{1}{4}-y,\, \frac{1}{4}-x,\, \frac{1}{4}-z\right]$	[8,40]
9	$[-x,\, -y,\, -z]$	[9,41]
10	$\left[y+\frac{3}{4},\, \frac{1}{4}-x,\, \frac{3}{4}-z\right]$	[10,42]
11	$\left[\frac{1}{4}-y,\, x+\frac{1}{4},\, \frac{3}{4}-z\right]$	[11,43]
12	$\left[-x,\, y,\, z+\frac{1}{2}\right]$	[12,44]
13	$\left[x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[13,45]
14	$\left[x,\, y+\frac{1}{2},\, -z\right]$	[14,46]
15	$\left[\frac{3}{4}-y,\, \frac{1}{4}-x,\, z+\frac{1}{4}\right]$	[15,47]
16	$\left[y+\frac{1}{4},\, x+\frac{1}{4},\, z+\frac{1}{4}\right]$	[16,48]
17	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[17,49]
18	$\left[\frac{3}{4}-y,\, x+\frac{1}{4},\, z+\frac{3}{4}\right]$	[18,50]
19	$\left[y+\frac{1}{4},\, \frac{1}{4}-x,\, z+\frac{3}{4}\right]$	[19,51]
20	$\left[x,\, -y,\, \frac{1}{2}-z\right]$	[20,52]
21	$\left[-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[21,53]
22	$\left[\frac{1}{2}-x,\, -y,\, z+\frac{1}{2}\right]$	[22,54]
23	$\left[y+\frac{1}{4},\, x+\frac{3}{4},\, \frac{3}{4}-z\right]$	[23,55]
24	$\left[\frac{3}{4}-y,\, \frac{3}{4}-x,\, \frac{3}{4}-z\right]$	[24,56]
25	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[25,57]
26	$\left[y+\frac{1}{4},\, \frac{3}{4}-x,\, \frac{1}{4}-z\right]$	[26,58]
27	$\left[\frac{3}{4}-y,\, x+\frac{3}{4},\, \frac{1}{4}-z\right]$	[27,59]
28	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$	[28,60]
29	$\left[x+\frac{1}{2},\, -y,\, z\right]$	[29,61]

continued ...

Table 7

No.	position	mapping
30	$\left[x+\frac{1}{2},\, y,\, \frac{1}{2}-z\right]$	[30,62]
31	$\left[\frac{1}{4}-y,\, \frac{3}{4}-x,\, z+\frac{3}{4}\right]$	[31,63]
32	$\left[y+\frac{3}{4},\, x+\frac{3}{4},\, z+\frac{3}{4}\right]$	[32,64]